Exclusion of a finite prime-covering period in Erdős Problem #203

Filip Majchrzak

Independent researcher, Poland

Research draft v1.1 — 8 November 2026 — not submitted or peer reviewed

(v1.0: 1 November 2026; changes listed before the references)

Abstract

Erdős Problem #203 asks whether some positive integer m, coprime to 6, makes every $2^k 3^\ell m + 1$ composite for $k, \ell \geq 0$. We rule out a specific finite-prime-covering approach: no *pure* covering by prime-divisibility cosets has covering period dividing $N = \operatorname{lcm}(1, \ldots, 17) = 12\,252\,240$. A complete computation of the relevant prime pool gives 192 distinct primes. Combining the known slope-one line-projection method with explicitly specified nonnegative weights on residue classes modulo 60 yields the exact upper bound $3\,961\,943/4\,084\,080 < 1$, contradicting the intensity required by a covering. The result extends immediately to all 480 divisors of N by inclusion of prime pools; 411 of these periods were excluded in earlier work, and to our knowledge the other 69 had not been excluded before. Exact-arithmetic verifiers, including a GMP recomputation of the 2,432-bit pool gcd, are publicly archived, and the pool and the bound have been reproduced by Erdős Problem a Day, a separate project (private communication). A rational linear-programming dual certificate additionally proves optimality of the weights *within this fixed cut and modulus*. The full Erdős problem remains open.

Keywords: covering systems; multiplicative orders; finite cyclic groups; line projection; computational number theory; exact certificates.

1 Problem and precise scope

Erdős and Graham asked whether there is an integer $m \geq 1$ with $\gcd(m, 6) = 1$ such that

$$2^k 3^\ell m + 1 \text{ is composite for every } k, \ell \geq 0. \tag{1}$$

This is Erdős Problem #203 [1, 2]; the unrestricted problem remains open. We consider only the *pure finite prime-covering* route, with no additional algebraic factorisations from special choices of m.

For each prime $p > 3$, define

$$H_p = \langle 2, 3 \rangle \subseteq \mathbb{F}_p^\times, \qquad e_p = |H_p| = \operatorname{lcm}\big(\operatorname{ord}_p(2), \operatorname{ord}_p(3)\big), \tag{2}$$

and the homomorphism $\phi_p : \mathbb{Z}^2 \to H_p$, $(k, \ell) \mapsto 2^k 3^\ell$. Its kernel L_p has index e_p. If p divides some $2^k 3^\ell m + 1$, the set of exponent pairs at which p divides this expression is one coset of L_p. (Otherwise the set is empty; this happens exactly when $p \mid m$, or $p \nmid m$ and $-m^{-1} \notin H_p$.) The formula for $|H_p|$ in (2) and the density condition $\sum 1/e_p \geq 1$ for finite coverings were stated with proofs on the discussion thread of the problem [3]; both are elementary.

A *pure finite prime covering* is a finite family of such cosets, with at most one coset per distinct prime, whose union covers $\mathbb{Z}^2$. "Pure" means that only prime divisibility is used: no algebraic factorisations, such as those available when m is a perfect power. Period T means the least common multiple of the indices e_p of the primes used. Only primes $p > 3$ need to be considered. For $\gcd(m, 6) = 1$, the prime 2 divides $2^k 3^\ell m + 1$ only when $k = 0$, and 3 only when $\ell = 0$; so the cosets of primes $p > 3$ must cover all pairs with $k, \ell \geq 1$. Since the family is finite, all

these cosets are invariant under translation by $T\mathbb{Z}^2$, so they then cover $\mathbb{Z}^2$. Hence, for such finite families, covering $\mathbb{Z}^2$, covering all nonnegative exponent pairs and covering all pairs with $k, \ell \geq 1$ are equivalent. This formulation is a necessary condition for a divisibility-only covering construction, not a characterisation of all possible solutions of (1).

Put $N = 12\,252\,240 = \operatorname{lcm}(1, \ldots, 17)$, and

$$\mathcal{P}_N = \{p > 3 \text{ prime} : e_p \mid N\}. \tag{3}$$

Theorem 1. *No pure finite prime covering for Erdős Problem #203 has period dividing* $N = 12\,252\,240$.

The proof combines the slope-one line restriction developed in prior work [5] with a new exact weighted certificate for this larger prime pool. It does not assert that no integer m satisfying (1) exists.

2 A finite line-projection obstruction

The following elementary fact is included for self-containment. The restriction of a coset cover to the cosets of a subgroup is classical [6, 7, 8]; our application uses it on lines of direction $(1, 1)$, following [5].

Lemma 1 (Density on a slope-one line)**.** *Let* $p > 3$ *be prime, and set*

$$h_p = \operatorname{ord}_p(6), \qquad i_p = e_p/h_p, \qquad d_p = \gcd(i_p, 60).$$

For each coset C_p *of* L_p*, there exists* $a_p \pmod{i_p}$ *such that, on the diagonal line* $k - \ell = \tau$*, the proportion of exponent pairs in* C_p *is*

$$\rho_p(\tau) = \begin{cases} i_p/e_p, & \tau \equiv a_p \pmod{i_p}, \\ 0, & \textit{otherwise.} \end{cases}$$

When these lines are pooled by $\tau \pmod{60}$*, the mean proportion contributed by* p *to class* t *is*

$$\overline{\rho}_p(t) = \begin{cases} d_p/e_p, & t \equiv a_p \pmod{d_p}, \\ 0, & \textit{otherwise.} \end{cases} \tag{4}$$

Proof. Every point with $k - \ell = \tau$ is of the form $(\tau + u, u)$; hence its image under ϕ_p is $2^\tau 6^u$. The quotient $H_p/\langle 6\rangle$ is cyclic of order i_p, generated by $2\langle 6\rangle$, since $\langle 2, 3\rangle = \langle 2, 6\rangle$. Membership in a fixed coset of L_p is therefore possible on precisely one class $\tau \equiv a_p \pmod{i_p}$. On such a line, exactly one u-class modulo h_p satisfies it, yielding proportion $1/h_p = i_p/e_p$. Inside a fixed class $t \pmod{60}$, the congruence $\tau \equiv a_p \pmod{i_p}$ has relative density d_p/i_p if $t \equiv a_p \pmod{d_p}$, and 0 otherwise. Multiplying by i_p/e_p proves (4). All densities can equivalently be calculated on the finite torus $(\mathbb{Z}/N\mathbb{Z})^2$, because the relevant orders divide N. □

If 5 or 7 does not occur in the covering, first add an arbitrary coset of L_5 (resp. L_7); this only enlarges the union, so a covering remains a covering, and the period still divides N. For $p = 5$, $e_5 = i_5 = 4$ and $d_5 = 4$; for $p = 7$, $e_7 = 6$, $h_7 = 2$, $i_7 = d_7 = 3$. A single global translation of the exponent lattice moves both selected residues $a_5 \pmod 4$ and $a_7 \pmod 3$ to 0 simultaneously, by the Chinese remainder theorem. We may hence normalise

$$a_5 \equiv 0 \pmod 4, \qquad a_7 \equiv 0 \pmod 3. \tag{5}$$

This translation does not affect whether the cosets cover the exponent lattice.

Lemma 2 (Weighted line obstruction)**.** *Let $(\lambda_t)_{t\in\mathbb{Z}/60\mathbb{Z}}$ be nonnegative rational weights with $\sum_t \lambda_t = 1$, supported on classes t with $t \not\equiv 0 \pmod 3$ and $t \not\equiv 0 \pmod 4$. For each $d \mid 60$ put*

$$\mu_d = \max_{c \pmod d} \sum_{\substack{t \pmod{60} \\ t\equiv c \ (d)}} \lambda_t, \qquad R_d = \sum_{\substack{p\in\mathcal{P}_N\setminus\{5,7\} \\ d_p=d}} \frac{1}{e_p}.$$

If

$$B(\lambda) := \sum_{d\mid 60} d\,\mu_d R_d < 1, \tag{6}$$

then the cosets of primes from $\mathcal{P}_N$ cannot cover $\mathbb{Z}^2$.

Proof. Suppose they cover. On each line $k-\ell=\tau$, the sum of coset densities is at least one by the union bound. The same is true after averaging over the lines $\tau \equiv t \pmod{60}$; this average is a finite average on the torus $(\mathbb{Z}/N\mathbb{Z})^2$, as remarked at the end of the proof of Lemma 1, because 60 and every e_p divide N. Take the weighted mean over t using λ_t. The normalised primes $5, 7$ contribute zero because the support avoids their classes in (5). By Lemma 1, each other prime p contributes at most $(d_p/e_p)\mu_{d_p}$, whatever residue a_p it selects. Thus the weighted sum is both at least 1 and at most $B(\lambda)$, contradicting (6). □

3 Complete finite prime pool and exact certificate

The crucial finiteness reduction, the gcd method already used for complete pools in [3, 4], is

$$p \in \mathcal{P}_N \iff p \mid \gcd(2^N-1, 3^N-1) =: G_N \qquad (p>3 \text{ prime}). \tag{7}$$

Indeed $e_p \mid N$ is equivalent to $2^N \equiv 3^N \equiv 1 \pmod p$. A provided factorisation into primes is therefore *complete* once the factors are proved prime and their product is checked to equal G_N.

Our data contain 192 distinct primes, the largest being 8 168 161. Exactly $5, 7, 11, 13, 17$ have valuation 2 in G_N; the remaining 187 have valuation 1. The full factor product has **2,432 bits and exactly 732 decimal digits**. The last number is corrected from the initial frozen report, which had used a GMP base-size estimate rather than the exact length of its decimal expansion. A GMP computation verifies the equality between the product and the directly calculated gcd, and a separate Python checker verifies primality and all multiplicative orders; both are part of the public verification package (Section 5).

Choose the explicit weights

$$\lambda_t = \begin{cases} 1/20, & t \in \{1, 26, 41, 46\}, \\ 1/10, & t \in \{2, 5, 7, 10, 14, 19, 23, 58\}, \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

They are nonnegative, sum to 1, and vanish whenever $t \equiv 0 \pmod 3$ or $t \equiv 0 \pmod 4$.

For transparency, Table 1 displays the entire grouped exact-rational calculation. The last column is the contribution $N\,d\mu_d R_d$; the relevant nonzero values of d are $1, 2, 3, 4, 5, 6, 10, 12, 15, 20$.

In particular,

$$\boxed{B = \frac{11\,885\,829}{12\,252\,240} = \frac{3\,961\,943}{4\,084\,080} < 1, \qquad 1 - B = \frac{122\,137}{4\,084\,080}.} \tag{9}$$

This and Lemma 2 prove that the full N-pool does not admit a pure covering.

Table 1: The exact weighted obstruction grouped by $d = \gcd(i_p, 60)$ (excluding primes 5 and 7).

d	μ_d	R_d	$Nd\mu_d R_d$
1	1	319967/556920	7,039,274
2	1/2	53617/291720	2,251,914
3	1/2	120079/2042040	1,080,711
4	1/2	91723/4084080	550,338
5	1/5	92789/2450448	463,945
6	1/4	10343/875160	217,203
10	1/10	16/36465	5,376
12	1/4	181/32760	203,082
15	1/10	61/16380	68,442
20	1/10	1/4420	5,544
		Total NB	**11,885,829**

Proof of Theorem 1. If a pure covering has period $T \mid N$, every one of its primes has $e_p \mid T$ and hence lies in $\mathcal{P}_N$. The same nonnegative weights and the same bound for the *full* pool apply, and deleting any primes cannot increase its upper bound. Equation (9) is already below 1. Consequently no such covering can exist. □

The divisor-count refinement in the earlier experimental report is unnecessary for this argument: since $N = 2^4 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17$, it has $5 \cdot 3 \cdot 2^5 = 480$ positive divisors, all covered at once by monotonicity. This does *not* assert that each of these 480 periods is a previously unknown exclusion. Earlier work excluded every period up to 65 520 [4] (with a second proof in [5]) and, among other periods, every divisor of $720\,720 = \mathrm{lcm}(1, \ldots, 16)$ [5]; its other divisor families, such as 110 880 and 332 640, contribute no further divisors of N. This covers 411 of the 480 divisors of N. The remaining 69 divisors, from 67 320 up to N, are all multiples of 17, and as far as we know none of them had been excluded before. For $T \mid N$ write $\mathcal{P}_T = \{p \in \mathcal{P}_N : e_p \mid T\}$. Once these pools are known, most of the new periods need only the earlier tools (Table 2).

Table 2: The 69 new divisors T of N, by the weakest argument that suffices.

Number of periods	Weakest sufficient argument
50	the pool density $\sum_{p \in \mathcal{P}_T} 1/e_p$ is below 1
18	the 5–7 cut with uniform weights gives a bound below 1 (largest bound 0.9092, at $T = 942\,480$)
1 ($T = N$)	uniform weights fail; non-uniform weights such as (8) give a bound below 1

Uniform weights $\lambda_t = 1/30$ on the 30 classes $t \pmod{60}$ with $3 \nmid t$ and $4 \nmid t$ (the set S of Section 4) give $B(\lambda) = 1\,557\,209/1\,531\,530 > 1$ for the full N-pool, so within this cut non-uniform weights, such as (8), are needed for N. The list of the 69 periods is part of the verification package.

4 Optimality within the fixed weighted cut

Finding the rational weights in (8) used computational optimisation, but the obstruction itself needs no optimiser. Here we can prove an additional, narrowly scoped fact.

Let $S = \{t \pmod{60} : 3 \nmid t,\ 4 \nmid t\}$, so $|S| = 30$. Retain the primes 5, 7 normalised as above, the slope-one line direction and modulus 60. Over all weights $\lambda_t \geq 0$ supported on S, minimising

$B(\lambda)$ is the linear program

$$\begin{aligned} &\text{minimise } \sum_d dR_d z_d, \\ &\text{subject to } \sum_{t\in S} \lambda_t = 1, \quad \lambda_t \geq 0, \\ &\qquad z_d \geq \sum_{t\equiv c \pmod d} \lambda_t \quad (\text{all } d \text{ with } R_d > 0,\ c \pmod d). \end{aligned} \tag{10}$$

Its dual admits nonnegative rational values $q_{d,c}$ and a lower bound L whenever

$$\sum_{c \pmod d} q_{d,c} \leq dR_d \quad (\forall d), \qquad \sum_d q_{d,t \bmod d} \geq L \quad (\forall t \in S). \tag{11}$$

Indeed, multiplying the first constraints by primal variables or averaging the second ones against λ shows $B(\lambda) \geq L$ for every admissible λ.

Proposition 1 (Optimality for the fixed cut)**.** *For the line direction* $(1,1)$*, normalised 5–7 support and weights on* $S \pmod{60}$*, the minimum of* $B(\lambda)$ *equals* $3\,961\,943/4\,084\,080$*.*

Proof. The weights (8) give the upper bound in (9). Appendix A supplies an explicit sparse rational assignment $q_{d,c}$ satisfying (11) with $L = 3\,961\,943/4\,084\,080$. Every inequality is checked using rational arithmetic in the companion verifier. Hence weak duality gives the matching lower bound. □

This claim of optimality is *not* a claim of optimality among all possible directions, moduli or prime-covering obstructions.

5 Verification, prior work and limitations

The verification package is archived at doi:10.5281/zenodo.24050485 and available at `https://github.com/kosza123/erdos203-period-12252240`. It contains (i) the complete factor list of G_N and the per-prime data, (ii) a standard-library-only checker of primality, orders, weights, the exact bound, the dual certificate and the divisors, (iii) a checker that recomputes G_N from 2^N-1 and 3^N-1 with GMP and compares it with the factor list, (iv) a from-scratch check that starts from N alone, finds the pool by brute force up to $3\cdot 10^7$, proves completeness with GMP and verifies Lemma 1 by direct enumeration for all cosets of the 38 pool primes with $e_p \leq 260$, (v) a script listing the 69 new periods, and (vi) the outputs of the verifiers on Windows and Linux, with a dated record of every documented check. Integer arithmetic and exact fractions, rather than decimal approximations, determine every asserted inequality. The optimisation routine is not needed to rerun the proof.

Within the author's AI-assisted workflow, the computation has also been reproduced by several separate implementations, written by separate Claude agent instances without access to the earlier checkers or to each other's code (described in `results/VERIFICATION.md`); they agree exactly on the pool (by direct gcd with GMP and with plain CPython integers, and, as a cross-check for primes below 2^{32}, by a sieve), the per-prime data, the bound B and the dual certificate. Erdős Problem a Day (Patrick White and Claude), a separate project, reviewed the submission on which version 1.0 is based, recomputed the pool and the bound with its own PARI/GP and plain-Python code, obtaining results identical to ours, and reported the limited claim of Theorem 1 as verified (private communication, 1 November 2026). These are checks of the computation and the argument; they are *not* a substitute for professional mathematical peer review.

The setup and the method come from earlier public work. On the discussion thread of the problem [3], AnimishSharma stated and proved the formula for $|H_p|$ and the density condition $\sum 1/e_p \geq 1$ (both elementary) and used the latter to exclude all periods below 5040 among primes up to $5 \cdot 10^7$; veljjanoski formulated the problem as a covering by cosets of lattices of index e_p and computed complete prime pools through gcds of cyclotomic values (all primes with $e_p \leq 5000$). Erdős Problem a Day excluded all periods up to 65 520 [4]; its later entry [5] introduced the weighted condition on lines and the slope-one 5–7 cut with uniform weights modulo 60, gave a second proof for these periods and excluded, among others, all divisors of 720 720. It credits the underlying restriction of a coset cover to the cosets of a subgroup to B. H. Neumann [6], M. J. Tomkinson [7] and Z.-W. Sun [8]. The present computation extends the explicit obstruction to $N = \mathrm{lcm}(1, \ldots, 17)$, which that account identified as the next frontier; what is new is the complete pool of N, from which the pools of the 69 new divisors are read off (the pools of all periods up to 65 520 and of the divisors of 720 720 had already been computed [3, 4, 5]), and explicit non-uniform weights such as (8). A preliminary literature search is *not* a proof of exhaustive historical novelty. The mathematical argument above and the included exact verifiers are intended to stand on their own.

The author used an AI-assisted research workflow, including SAPIO MORPHEUS/COMPLEX and ChatGPT, for algorithm design, computation, checking and preparation of version 1.0 of this draft. Version 1.1 was revised, and the verification package prepared and re-checked, with Claude (Anthropic) acting as a research operator under the author's direction. The separate implementations mentioned above were written by separate Claude agent instances, whose review also prompted most of the expository changes listed below. All mathematical claims are explicitly limited to the certified class and remain the author's responsibility. We claim neither a full solution to Erdős #203 nor any conclusion about non-pure or nonperiodic mechanisms.

Acknowledgements

The author thanks Erdős Problem a Day (Patrick White and Claude) for an independent check of the submission on which version 1.0 is based (1 November 2026) and for comments on its exposition and attribution, which are reflected in versions 1.0 and 1.1.

Changes from version 1.0

Version 1.1 defines "pure" explicitly; states exactly when a prime contributes no coset; explains why only primes $p > 3$ are needed, how an unused 5 or 7 is handled, and why the averaging over lines is finite; states which of the 480 divisors of N are new, with Table 2 and the uniform-weight bound for N; credits the forum contributions, both earlier Erdős Problem a Day entries and the classical lemma; describes the public verification package and the reproductions, including the check by Erdős Problem a Day; updates the statement on AI use; adds acknowledgements; gives package paths in Appendix A; and no longer calls the author's own checks "independent". The theorem, the weights, the bound and the dual certificate are unchanged.

References

[1] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographie de L'Enseignement Mathématique 28, Université de Genève (1980), p. 27.

[2] T. F. Bloom (ed.), "Erdős Problem #203," https://www.erdosproblems.com/203 (accessed 8 November 2026).

[3] Erdős Problems, "203 Discussion Thread" (contributions by AnimishSharma, 21 June 2026, and veljjanoski, 15 September 2026), https://www.erdosproblems.com/forum/thread/203 (accessed 8 November 2026).

[4] Erdős Problem a Day (Patrick White and Claude), "Erdős #203: the cosets of 5 and 7 always overlap — no covering system has period $\leq 65\,520$," 12 October 2026, https://erdosproblemaday.com/day/203-no-covering-system-with-period-up-to-30000 (accessed 8 November 2026).

[5] Erdős Problem a Day (Patrick White and Claude), "Erdős #203: along the lines of slope 1 the primes 5 and 7 leave half the lines to the rest of the pool," 14 October 2026, https://erdosproblemaday.com/day/203-the-quotient-lemma-lines-close-the-open-pools (accessed 8 November 2026).

[6] B. H. Neumann, Groups covered by finitely many cosets, *Publ. Math. Debrecen* 3 (1954), 227–242.

[7] M. J. Tomkinson, Groups covered by finitely many cosets or subgroups, *Comm. Algebra* 15 (1987), no. 4, 845 ff.

[8] Z.-W. Sun, Finite covers of groups by cosets or subgroups, *Internat. J. Math.* 17 (2006), 1047–1064.

A An explicit dual certificate

For completeness, the following are the *nonzero* entries $q_{d,c}$ of a feasible dual solution for (11). Unlisted entries vanish. Set $L = 3\,961\,943/4\,084\,080$. The companion file `data/optional_lp_dual.json` and the Python standard-library script `verify/verify_dual_stdlib.py` in the verification package check all $\sum_c q_{d,c} \leq dR_d$ constraints and all 30 inequalities $\sum_d q_{d,t \bmod d} \geq L$ exactly, with no floating point operations.

d	c	$q_{d,c}$	d	c	$q_{d,c}$
1	0	319967/556920	10	0	161/58344
2	0	16519/157080	10	5	95/58344
2	1	31523/120120	12	10	181/2730
3	1	27697/291720	15	2	61/5460
3	2	83179/1021020	15	5	61/5460
4	2	91723/1021020	15	8	61/5460
5	0	89597/2450448	15	11	61/5460
5	1	7199/188496	15	14	61/5460
5	2	7199/188496	20	2	1/884
5	3	7199/188496	20	6	1/884
5	4	7199/188496	20	14	1/884
6	2	28019/408408	20	18	1/884
6	5	1569/680680			

Thus the proof of Proposition 1 can be checked by addition and comparison of explicit rational numbers. The main theorem does not depend on this proposition.